

Chapter 10

Penalty Function Methods

10.1 Penalty Function

The penalty function methods are an important class of methods for constrained optimization problem

$$\min_{x \in R^n} f(x) \quad (10.1.1)$$

$$\text{s.t.} \quad c_i(x) = 0, \quad i \in E \stackrel{Def}{=} \{1, \dots, m_e\}, \quad (10.1.2)$$

$$c_i(x) \geq 0, \quad i \in I \stackrel{Def}{=} \{m_e + 1, \dots, m\}. \quad (10.1.3)$$

In this class of methods we replace the original constrained problem by a sequence of unconstrained subproblems that minimizes the penalty functions. The penalty function is a function with penalty property

$$P(x) = \bar{P}(f(x), c(x)), \quad (10.1.4)$$

constructed from the objective function $f(x)$ and the constraints $c(x)$. The so-called “penalty” property requires $P(x) = f(x)$ for all feasible points $x \in X$ of (10.1.1)–(10.1.3), and $P(x)$ is much larger than $f(x)$ when the constraint violations are severe.

To describe the degree of constraint violation, we define the constraint violation function $c^{(-)}(x) = (c_1^{(-)}(x), \dots, c_m^{(-)}(x))^T$ as follows:

$$c_i^{(-)}(x) = c_i(x), \quad i \in E, \quad (10.1.5)$$

$$c_i^{(-)}(x) = \min\{c_i(x), 0\}, \quad i \in I. \quad (10.1.6)$$

Define

$$C = \{c_i(x) \mid c_i(x) = 0, i \in E; c_i(x) \geq 0, i \in I\}. \quad (10.1.7)$$

Obviously, x is a feasible point if and only if $c(x) \in C$. Furthermore,

$$\text{if } c_i(x) \geq 0, \text{ i.e., } c_i(x) \in C, \text{ then } c_i^{(-)}(x) = 0;$$

$$\text{if } c_i(x) < 0, \text{ i.e., } c_i(x) \notin C, \text{ then } c_i^{(-)}(x) \neq 0.$$

This means, for each constraint, the constraint violation function is nonzero when the corresponding constraint is violated and zero when the corresponding constraint is feasible.

It is not difficult to see that for any $x \in R^n$, we have

$$\|c^{(-)}(x)\|_2 = \text{dist}(c(x), C), \quad (10.1.8)$$

where $\text{dist}(\cdot, \cdot)$ denotes the distance from a point to a set and is defined as

$$\text{dist}(x, Y) = \min\{\|x - y\|_2 \mid \forall y \in Y\}. \quad (10.1.9)$$

The penalty function consists of a sum of the original objective function and a penalty term, i.e.,

$$P(x) = f(x) + h(c^{(-)}(x)), \quad (10.1.10)$$

where the penalty term $h(c^{(-)}(x))$ is a function defined on R^m and satisfies

$$h(0) = 0, \quad \lim_{\|c\| \rightarrow +\infty} h(c) = +\infty. \quad (10.1.11)$$

The earliest penalty function is the Courant penalty function, or called the quadratic penalty function, defined as

$$P(x) = f(x) + \sigma \|c^{(-)}(x)\|_2^2, \quad (10.1.12)$$

where $\sigma > 0$ is a positive constant, which is called the penalty parameter. We give an example to describe the penalty function.

$$\begin{aligned} \min \quad & x \\ \text{s.t.} \quad & x - 2 \geq 0. \end{aligned} \quad (10.1.13)$$

Then

$$h(c^{(-)}(x)) = \|c^{(-)}(x)\|_2^2 = [\min\{0, x - 2\}]^2 = \begin{cases} 0 & \text{if } x \geq 2, \\ (x - 2)^2 & \text{if } x < 2. \end{cases}$$

Note that the minimum of $f(x) + \sigma\|c^{(-)}(x)\|^2$ occurs at the point $2 - \frac{1}{\sigma}$, and approaches the minimum point $\bar{x} = 2$ of the original problem, as σ approaches ∞ .

Obviously, (10.1.12) is a particular case of (10.1.10) in which $h(c) = \sigma\|c\|_2^2$. In fact, for any norm on R^m and any $\alpha > 0$, the function $h(c) = \sigma\|c\|^\alpha$ satisfies (10.1.11). So, a class of penalty functions can be defined as:

$$P(x) = f(x) + \sigma\|c^{(-)}(x)\|^\alpha, \quad (10.1.14)$$

where $\sigma > 0$ is a penalty parameter, $\alpha > 0$, and $\|\cdot\|$ is some norm on R^m . Typically, (10.1.12) is often written as

$$P(x) = f(x) + \frac{1}{2}\sigma\|c^{(-)}(x)\|^2 \quad (10.1.15)$$

$$= f(x) + \frac{1}{2}\sigma \sum_{i=1}^{m_e} c_i^2(x) + \frac{1}{2}\sigma \sum_{i=m_e+1}^m [c_i^{(-)}(x)]^2 \quad (10.1.16)$$

and is called the quadratic penalty function, where $\sigma > 0$ and $c_i^{(-)}(x) = \min\{0, c_i(x)\}$.

Besides (10.1.12), the common particular forms of (10.1.14) are

$$P_1(x) = f(x) + \sigma\|c^{(-)}(x)\|_1 \quad (10.1.17)$$

and

$$P_\infty(x) = f(x) + \sigma\|c^{(-)}(x)\|_\infty, \quad (10.1.18)$$

which are called L_1 penalty function and L_∞ penalty function respectively.

If the penalty function takes values approaching $+\infty$ as x approaches the boundary of the feasible region, it is called the interior point penalty function. The interior point penalty function is suitable only to inequality-constrained problems, i.e., $m_e = 0$. Typically, the two most important interior point penalty functions are the inverse barrier function

$$P(x) = f(x) + \frac{1}{\sigma} \sum_{i=1}^m \frac{1}{c_i(x)} \quad (10.1.19)$$

and the logarithmic barrier function

$$P(x) = f(x) - \frac{1}{\sigma} \sum_{i=1}^m \log c_i(x). \quad (10.1.20)$$

If given an initial point in the interior of the feasible region, the whole sequence generated by the interior point penalty function method is interior points. Since these functions set an infinitely high “barrier” on the boundary, they are also said to be barrier functions.

Let x^* be a KKT point of constrained optimization (10.1.1)–(10.1.3). Then it follows from (10.1.12) that $\nabla P(x^*) = \nabla f(x^*)$. In general, x^* is not a stationary point of the Courant penalty function, and the penalty function method attempts to create a local minimizer at x^* in the limit $\sigma_k \rightarrow \infty$. To overcome this shortcoming, we introduce parameters θ_i ($i = 1, \dots, m$) with $\theta_i \geq 0$ ($i = m_e + 1, \dots, m$) to change the origin of the penalty term. Write $\theta = (\theta_1, \dots, \theta_m)^T$. Modifying (10.1.12) gives

$$\begin{aligned} P(x) &= f(x) + \sum_{i=1}^m \frac{\sigma_i}{2} \left([c_i(x) - \theta_i]_-^2 - \theta_i^2 \right) \\ &= f(x) + \sum_{i=1}^{m_e} \left[-\lambda_i c_i(x) + \frac{1}{2} \sigma_i (c_i(x))^2 \right] \\ &\quad + \sum_{i=m_e+1}^m \begin{cases} -\lambda_i c_i(x) + \frac{1}{2} \sigma_i (c_i(x))^2, & \text{if } c_i(x) < \frac{\lambda_i}{\sigma_i}; \\ -\frac{1}{2} \lambda_i^2 / \sigma_i, & \text{otherwise} \end{cases} \end{aligned} \quad (10.1.21)$$

where

$$\lambda_i = \sigma_i \theta_i, \quad i = 1, \dots, m. \quad (10.1.22)$$

Because the penalty function (10.1.21) can be obtained from Lagrange function (8.2.18) by adding a penalty term, (10.1.21) is referred to as an augmented Lagrangian function. Alternatively, (10.1.21) can be also obtained from the penalty function (10.1.12) by adding a multiplier term $-\lambda^T c$, (10.1.21) is also called a multiplier penalty function. Let x^* be a KKT point of the constrained optimization problem, and λ_i^* ($i = 1, \dots, m$) corresponding Lagrange multipliers. Then the augmented Lagrangian function with λ_i^* ($i = 1, \dots, m$) satisfies $\nabla P(x^*) = 0$. In addition, the Lagrange multiplier λ^* is not known in advance, so the augmented Lagrangian function method needs to update λ_i ($i = 1, \dots, m$) successfully.

For equality-constrained problem ($m_e = m$), we define

$$\lambda(x) = (A(x))^+ g(x), \quad (10.1.23)$$

where $A(x) = (\nabla c_1(x), \dots, \nabla c_m(x))$ is an $n \times m$ matrix, $g(x) = \nabla f(x)$, A^+ denotes the generalized inverse of A , and the multiplier $\lambda(x)$ is the minimum l_2 norm solution of the least-squares problem

$$\min_{\lambda \in R^m} \left\| \nabla f(x) - \sum_{i=1}^m \lambda_i \nabla c_i(x) \right\|_2^2. \quad (10.1.24)$$

By using (10.1.23), we can give Fletcher's smooth exact penalty function (or Fletcher's augmented Lagrangian function) for the case where only equality constraints are present in (10.1.1)–(10.1.3) as follows:

$$P(x) = f(x) - \lambda(x)^T c(x) + \frac{1}{2} \sum_{i=1}^m \sigma_i (c_i(x))^2, \quad (10.1.25)$$

where $\sigma_i > 0$ ($i = 1, \dots, m$) are penalty parameters.

Let x^* be the solution of the equality-constrained problem, $A(x^*)$ have full column rank,

$$\begin{aligned} \nabla_x P(x^*) &= g(x^*) - A(x^*)\lambda^* = 0, \\ \nabla_{xx}^2 P(x^*) &= W^* + A(x^*)A(x^*)^+ W^* \\ &\quad + W^* A(x^*)A(x^*)^+ + A(x^*)DA(x^*)^T, \end{aligned} \quad (10.1.26)$$

where $D = \text{diag}(\sigma_1, \dots, \sigma_m)$ and

$$W^* = \nabla^2 f(x^*) - \sum_{i=1}^m \lambda_i^* \nabla^2 c_i(x^*) = \nabla_{xx}^2 L(x^*, \lambda^*).$$

Lemma 10.1.1 *Let $H \in R^{n \times n}$ be symmetric and $A \in R^{n \times m}$. If*

$$d^T H d > 0 \quad (10.1.27)$$

for any nonzero vector d with $A^T d = 0$, then there exists $\sigma \geq 0$ such that

$$H + \sigma A A^T \quad (10.1.28)$$

is a positive definite matrix.

Proof. By assumption, there is $\delta > 0$, such that if $\|A^T d\|_2 \leq \delta$ and $\|d\|_2 = 1$, then (10.1.27) holds. Consider

$$\min_{\substack{\|A^T d\|_2 \geq \delta \\ \|d\|_2 = 1}} \frac{d^T H d}{\|A^T d\|_2^2}. \quad (10.1.29)$$

Since the set $\{d \mid \|d\|_2 = 1, \|A^T d\|_2 \geq \delta\}$ is a finite and closed set, then the minimum of (10.1.29) is achieved. Hence there is η such that for all $d \in \{d \mid \|d\|_2 = 1, \|A^T d\|_2 \geq \delta\}$ we have

$$\frac{d^T H d}{\|A^T d\|_2^2} > \eta. \quad (10.1.30)$$

Set $\sigma = \max\{-\eta, 0\}$. Therefore, for any d with $\|d\|_2 = 1$, we have

$$d^T (H + \sigma A A^T) d > 0. \quad (10.1.31)$$

We complete the proof. $\square$

If the second-order sufficient condition

$$d^T W^* d > 0, \forall d \neq 0, (A^*)^T d = 0 \quad (10.1.32)$$

holds, then it follows from the above lemma that there exists $\bar{\sigma} \geq 0$ such that for all $\sigma_i \geq \bar{\sigma}$, the matrix $\nabla_{xx}^2 P(x^*)$ defined by (10.1.26) is positive definite. Therefore the penalty function (10.1.25) is said to be exact if the solution x^* of the original problem is also the strict local minimizer of the penalty function $P(x^*)$.

If Lagrange multipliers λ_i of the augmented Lagrangian function are the Lagrange multipliers λ_i^* at the solution x^* of the problem, then under the second-order sufficient condition (10.1.32), x^* is also the local minimizer of the augmented Lagrangian function (10.1.21) when σ is large enough. Thus, the augmented Lagrangian function is also an exact penalty function.

For an L_1 penalty function, if

$$\sigma > \|\lambda^*\|_\infty, \quad (10.1.33)$$

then under the second-order sufficient condition (10.1.32), the solution x^* of the original problem is also a strict local minimizer of the L_1 penalty function. Thus, the L_1 penalty function is referred to as an L_1 exact penalty function. Similarly, an L_∞ penalty function is also exact.

Note that the KKT point of the constrained optimization problem is not, in general, the stationary point of the Courant penalty function. Thus, the Courant penalty function is not an exact penalty function.

In this chapter, we will describe the simple penalty function method, interior point penalty function method (i.e., barrier function method), multiplier penalty function method, smooth exact penalty function method and non-smooth exact penalty function method.

10.2 The Simple Penalty Function Method

The penalty function method is an approach to minimize a sequence of penalty functions and obtain the minimizer of the original constrained optimization problem.

Consider the simple penalty function

$$P_\sigma(x) = f(x) + \sigma \|c^{(-)}(x)\|^\alpha, \quad (10.2.1)$$

where $\sigma > 0$ is the penalty parameter, $\alpha > 0$ a positive constant, and $\|\cdot\|$ a given norm on R^m . Write $x(\sigma)$ as a solution of problem

$$\min_{x \in R^n} P_\sigma(x). \quad (10.2.2)$$

Next, we first give some lemmas.

Lemma 10.2.1 *Let $0 < \sigma_1 < \sigma_2$. Then*

$$P_{\sigma_1}(x(\sigma_1)) \leq P_{\sigma_2}(x(\sigma_2)), \quad (10.2.3)$$

$$f(x(\sigma_1)) \leq f(x(\sigma_2)), \quad (10.2.4)$$

$$\|c^{(-)}(x(\sigma_1))\| \geq \|c^{(-)}(x(\sigma_2))\|. \quad (10.2.5)$$

Proof. From the definition of $x(\sigma)$, we have

$$P_{\sigma_1}(x(\sigma_1)) \leq P_{\sigma_1}(x(\sigma_2)) \leq P_{\sigma_2}(x(\sigma_2)) \leq P_{\sigma_2}(x(\sigma_1)), \quad (10.2.6)$$

which shows (10.2.3). By use of (10.2.6) and (10.2.1), we have

$$\begin{aligned} 0 &\leq P_{\sigma_1}(x(\sigma_2)) - P_{\sigma_2}(x(\sigma_2)) - [P_{\sigma_1}(x(\sigma_1)) - P_{\sigma_2}(x(\sigma_1))] \\ &= (\sigma_1 - \sigma_2) [\|c^{(-)}(x(\sigma_2))\|^\alpha - \|c^{(-)}(x(\sigma_1))\|^\alpha], \end{aligned} \quad (10.2.7)$$

which means that (10.2.5) holds. Using (10.2.6) and (10.2.5) gives

$$\begin{aligned} f(x(\sigma_1)) &\leq f(x(\sigma_2)) + \sigma_1 (\|c^{(-)}(x(\sigma_2))\|^\alpha - \|c^{(-)}(x(\sigma_1))\|^\alpha) \\ &\leq f(x(\sigma_2)). \end{aligned} \quad (10.2.8)$$

Hence (10.2.4) holds. We complete the proof. $\square$

Lemma 10.2.2 *Let $\delta = \|c^{(-)}(x(\sigma))\|$. Then $x(\sigma)$ is also the solution of problem*

$$\min_{x \in R^n} f(x) \quad (10.2.9)$$

$$\text{s.t.} \quad \|c^{(-)}(x)\| \leq \delta. \quad (10.2.10)$$

Proof. For any x satisfying (10.2.10), we have

$$\begin{aligned}
 0 &\leq \sigma(\|c^{(-)}(x(\sigma))\|^\alpha - \|c^{(-)}(x)\|^\alpha) \\
 &= P_\sigma(x(\sigma)) - f(x(\sigma)) - P_\sigma(x) + f(x) \\
 &= [P_\sigma(x(\sigma)) - P_\sigma(x)] + f(x) - f(x(\sigma)) \\
 &\leq f(x) - f(x(\sigma)).
 \end{aligned} \tag{10.2.11}$$

Hence, for any x satisfying (10.2.10), we have

$$f(x) \geq f(x(\sigma)), \tag{10.2.12}$$

which shows that $x(\sigma)$ is the solution of (10.2.9)–(10.2.10). $\square$

By the definition of the constraint violation function $c^{(-)}(x)$, the original problem (10.1.1)–(10.1.3) can be written equivalently as

$$\min_{x \in R^n} f(x), \tag{10.2.13}$$

$$\text{s.t.} \quad \|c^{(-)}(x)\| = 0. \tag{10.2.14}$$

Hence, if δ is sufficiently small, the problem (10.2.9)–(10.2.10) can be regarded as an approximation of (10.2.13)–(10.2.14), and so $x(\sigma)$ can be regarded as the approximate solution of the original problem. In fact, from Lemma 10.2.2, we know that when $c^{(-)}(x(\sigma)) = 0$, $x(\sigma)$ is just the solution of problem (10.1.1)–(10.1.3).

The basic idea of the penalty function method is that the penalty parameter σ is increased in each iteration until $\|c^{(-)}(x(\sigma))\|$ is smaller than a given tolerance. Below we give a penalty function method with the simple penalty function.

Algorithm 10.2.3 (*Simple Penalty Function Method*)

Step 1. Given $x_1 \in R^n, \sigma_1 > 0, \epsilon \geq 0, k := 1$.

Step 2. Find a solution $x(\sigma_k)$ of

$$\min_{x \in R^n} P_{\sigma_k}(x), \tag{10.2.15}$$

starting at x_k .

Step 3. If $\|c^{(-)}(x(\sigma_k))\| \leq \epsilon$, stop;

Set $x_{k+1} = x(\sigma_k), \sigma_{k+1} = 10\sigma_k$;

$k := k + 1$, turn to Step 2. $\square$

Note that the parameter $\{\sigma_k\}$ can be chosen flexibly and adoptively. It means that you can choose $\sigma_{k+1} = 10\sigma_k$ or $\sigma_{k+1} = 2\sigma_k$, which depends on the difficulty of minimizing the penalty function at iteration k .

Now we discuss the convergence property of Algorithm 10.2.3.

Theorem 10.2.4 *Suppose that the tolerance ϵ in Algorithm 10.2.3 satisfies*

$$\epsilon > \min_{x \in R^n} \|c^{(-)}(x)\|, \quad (10.2.16)$$

then the algorithm must terminate finitely.

Proof. Suppose, by contradiction, that the theorem is not true. Then there must exist $\sigma_k \rightarrow +\infty$ and for all k ,

$$\|c^{(-)}(x(\sigma_k))\| > \epsilon. \quad (10.2.17)$$

From (10.2.16), there exists $\hat{x} \in R^n$ such that

$$\|c^{(-)}(\hat{x})\| < \epsilon. \quad (10.2.18)$$

By use of the definition of $x(\sigma)$ and (10.2.4), we have

$$\begin{aligned} f(\hat{x}) + \sigma_k \|c^{(-)}(\hat{x})\|^\alpha &\geq f(x(\sigma_k)) + \sigma_k \|c^{(-)}(x(\sigma_k))\|^\alpha \\ &\geq f(x(\sigma_1)) + \sigma_k \|c^{(-)}(x(\sigma_k))\|^\alpha. \end{aligned} \quad (10.2.19)$$

By arranging (10.2.19) and taking the limit as $\sigma_k \rightarrow +\infty$, we obtain that

$$\begin{aligned} &\|c^{(-)}(\hat{x})\|^\alpha - \|c^{(-)}(x(\sigma_k))\|^\alpha \\ &\geq \frac{1}{\sigma_k} [f(x(\sigma_1)) - f(\hat{x})] \rightarrow 0, \end{aligned} \quad (10.2.20)$$

which contradicts (10.2.17)–(10.2.18). This completes the proof. $\square$

Theorem 10.2.5 *If Algorithm 10.2.3 does not terminate finitely, then*

$$\min_{x \in R^n} \|c^{(-)}(x)\| \geq \epsilon \quad (10.2.21)$$

and

$$\lim_{k \rightarrow \infty} \|c^{(-)}(x(\sigma_k))\| = \min_{x \in R^n} \|c^{(-)}(x)\|, \quad (10.2.22)$$

and any accumulation point x^ of $\{x(\sigma_k)\}$ is the solution of problem*

$$\min_{x \in R^n} f(x), \quad (10.2.23)$$

$$\text{s.t. } \|c^{(-)}(x)\| = \min_{y \in R^n} \|c^{(-)}(y)\|. \quad (10.2.24)$$

Proof. Suppose that the algorithm does not terminate finitely. It follows from Theorem 10.2.4 that (10.2.21) holds. Since $\sigma_k \rightarrow +\infty$ and by (10.2.20), we have that for given $\hat{x} \in R^n$,

$$\liminf_{k \rightarrow \infty} \left[\|c^{(-)}(\hat{x})\|^\alpha - \|c^{(-)}(x(\sigma_k))\|^\alpha \right] \geq 0 \quad (10.2.25)$$

which concludes (10.2.22).

Let x^* be any accumulation point of $\{x(\sigma_k)\}$. By (10.2.22), x^* must be feasible point of (10.3.25). If x^* is not the solution of (10.2.23)-(10.3.25), there exists $\bar{x}$ such that

$$f(\bar{x}) < f(x^*) \quad (10.2.26)$$

and

$$\|c^{(-)}(\bar{x})\| = \min_{y \in R^n} \|c^{(-)}(y)\|. \quad (10.2.27)$$

It follows from Lemma 10.2.1 that $f(x(\sigma_k))$ approach to $f(x^*)$. Then, by (10.2.26), we have that the inequality

$$f(\bar{x}) < f(x(\sigma_k)) \quad (10.2.28)$$

holds for k sufficiently large, which, together with (10.2.27), gives

$$\begin{aligned} f(\bar{x}) + \sigma_k \|c^{(-)}(\bar{x})\| &< f(x(\sigma_k)) + \sigma_k \min_y \|c^{(-)}(y)\| \\ &= f(x(\sigma_k)) + \sigma_k \|c^{(-)}(x(\sigma_k))\| \end{aligned} \quad (10.2.29)$$

for k sufficiently large. This is just

$$P_{\sigma_k}(\bar{x}) < P_{\sigma_k}(x(\sigma_k)). \quad (10.2.30)$$

This contradicts the definition of $x(\sigma_k)$. The contradiction proves our theorem. $\square$

The above two theorems establish a direct consequence.

Corollary 10.2.6 *Let problem (10.1.1)-(10.1.3) have feasible points. Then Algorithm 10.2.3 either finitely terminates at the solution of (10.2.9)-(10.2.10), or any accumulation points of the generated sequence are the solution of the original problem.*

For the Courant penalty function, i.e., $\|\cdot\|_2$ and $\alpha = 2$ in (10.2.1), we have

$$\nabla f(x(\sigma_k)) + 2\sigma_k \sum_{i=1}^m c_i^{(-)}(x(\sigma_k)) \nabla c_i^{(-)}(x(\sigma_k)) = 0. \quad (10.2.31)$$

Suppose that the infinite sequence $\{x_k\}$ from Algorithm 10.2.3 converges to x^* , we then have

$$\nabla f(x_{k+1}) = \sum_{i=1}^m \lambda_i^{(k+1)} \nabla c_i(x_{k+1}), \quad (10.2.32)$$

where

$$\lambda_i^{(k+1)} = -2\sigma_k c_i^{(-)}(x_{k+1}). \quad (10.2.33)$$

Hence the multiplier $\lambda^{(k+1)}$ given in (10.2.33) is an approximation to a Lagrangian multiplier. It is not difficult to see that if $x_k \rightarrow x^*$, $c^{(-)}(x^*) = 0$ and $\nabla c_i(x^*)$ ($i \in E \cup I(x^*)$) are linearly independent, then $\lambda_k \rightarrow \lambda^*$. Since, in the general case, $\|\lambda^*\|_2 \neq 0$, it follows from (10.2.33) that

$$\frac{1}{\sigma_k} = O(\|c^{(-)}(x_{k+1})\|_2) = O(\|x_{k+1} - x^*\|_2). \quad (10.2.34)$$

On the other hand, by using (10.2.32), $c^{(-)}(x^*) = 0$ and $\|\lambda^{k+1} - \lambda^*\| = O(\|x_{k+1} - x^*\|)$, we can obtain

$$\begin{bmatrix} W^* & -A^* \\ -(A^*)^T & 0 \end{bmatrix} \begin{pmatrix} x_{k+1} - x^* \\ \hat{\lambda}^{(k+1)} - \hat{\lambda}^* \end{pmatrix} = \begin{pmatrix} 0 \\ \hat{c}(x_{k+1}) \end{pmatrix} + O(\|x_{k+1} - x^*\|), \quad (10.2.35)$$

where

$$W^* = \nabla_{xx}^* L(x^*, \lambda^*), \quad (10.2.36)$$

A^* is a matrix consisting of $\nabla c_i(x^*)$ ($i \in E$ or $\lambda_i^* > 0$), $\hat{\lambda}^*$ is a vector consisting of those components of λ^* ($i \in E$ or $\lambda_i^* > 0$), and the definitions of $\hat{\lambda}^{k+1}$ and $\hat{c}(x_{k+1})$ are similar to $\hat{\lambda}^*$. By (10.2.33), we have

$$\|\hat{c}(x_{k+1})\| = O\left(\frac{1}{\sigma_k}\right). \quad (10.2.37)$$

If the second sufficient conditions (8.3.35)-(8.3.36) are satisfied, then the matrix

$$\begin{bmatrix} W^* & -A^* \\ -(A^*)^T & 0 \end{bmatrix} \quad (10.2.38)$$

is nonsingular. By use of (10.2.35) and (10.2.37), we have

$$\|x_{k+1} - x^*\| = O\left(\frac{1}{\sigma_k}\right). \quad (10.2.39)$$

Then, the above equality and (10.2.34) implies that the rate of $\|x_{k+1} - x^*\| \rightarrow 0$ is the same as that of $\frac{1}{\sigma_k} \rightarrow 0$. This phenomenon can be illustrated by the following example.

Example 10.2.7 Consider the problem

$$\min_{(x_1, x_2) \in R^2} \quad x_1 + x_2, \quad (10.2.40)$$

$$\text{s.t.} \quad x_2 - x_1^2 = 0. \quad (10.2.41)$$

Solution. For the Courant penalty function, we have

$$x(\sigma) = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{4} - \frac{1}{2\sigma} \end{bmatrix} = x^* - \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} \frac{1}{\sigma}, \quad (10.2.42)$$

where $x^* = (-1/2, 1/4)$ is the unique solution of (10.2.40)-(10.2.41). Thus, the sequence $\{x_k\}$ generated by Algorithm 10.2.3 satisfies

$$x_{k+1} - x^* = \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix} \frac{1}{\sigma_k}. \quad (10.2.43)$$

Therefore, we need to choose a very large penalty factor σ_k to solve constrained optimization problems. However, this leads to numerical difficulties of ill-conditioning. $\square$

If L_1 or L_∞ penalty functions are employed, in general, Algorithm 10.2.3 terminates finitely at an exact solution of the original problem. In fact, let x^* be a solution of the original constrained problem, and λ^* a corresponding Lagrange multiplier, then x^* is a minimizer of the L_1 exact penalty function if $\sigma_1 > \|\lambda^*\|_\infty$. Unfortunately, it is a nonsmooth optimization problem to minimize a L_1 exact penalty function. The topic about minimization of nonsmooth exact penalty function will be discussed in detail in §10.6.

10.3 Interior Point Penalty Functions

Similar to the penalty functions discussed in the previous sections, the interior point penalty functions are also used to transform a constrained problem

into an unconstrained problem or into a sequence of unconstrained problems. These functions set a barrier against leaving the feasible region, i.e., these functions are characterized by the property of preserving strict constraint feasibility at all times (i.e., the generated sequence always lies in the interior of the feasible region), by using an interior point penalty term which is infinite on the constraint boundaries. Such methods based on an interior point penalty function are referred to as interior point penalty function methods. Note that the interior point penalty function is also said to be a barrier function, and the corresponding techniques are known as barrier function methods.

The interior point function methods are used to deal with the inequality-constrained optimization problem

$$\min_{x \in R^n} f(x) \quad (10.3.1)$$

$$\text{s.t.} \quad c_i(x) \geq 0, \quad i = 1, \dots, m, \quad (10.3.2)$$

where

$$X = \{x \in R^n \mid c_i(x) \geq 0, \quad i = 1, 2, \dots, m\}$$

is a feasible region. The strictly feasible region is defined by

$$\text{int}(X) \triangleq \{x \in R^n \mid c_i(x) > 0 \text{ for all } i\}. \quad (10.3.3)$$

We assume that $\text{int}X$ is nonempty.

The interior point penalty function is of a general form

$$P_\sigma(x) = f(x) + \frac{1}{\sigma} \sum_{i=1}^m h(c_i(x)), \quad (10.3.4)$$

where $\sigma > 0$ is a barrier parameter, which controls the iteration. If $\{\sigma_k\} \rightarrow \infty$ is chosen, the barrier term becomes more and more negligible except close to the boundary. The $h(c_i)$ is a real function defined on $(0, +\infty)$ which satisfies that

$$\lim_{c_i \rightarrow 0^+} h(c_i) = +\infty, \quad (10.3.5)$$

which means the value $h(c_i)$ approaches ∞ as x approaches the boundary of $\text{int}(X)$, and that

$$h(c_1) \geq h(c_2), \quad \forall c_1 < c_2. \quad (10.3.6)$$

Some interior point penalty functions satisfy

$$h(c_i) > 0, \quad \forall c_i > 0. \quad (10.3.7)$$

As we have seen, the inverse barrier function (10.1.19) and the logarithmic barrier function (10.3.16) are the two most important special cases of (10.3.4). For the inverse barrier function, (10.3.7) holds.

Let $x(\sigma)$ be the solution of the problem

$$\min_{x \in R^n} P_\sigma(x). \quad (10.3.8)$$

Assume that we solve (10.3.8) with a strict interior point as an initial point. Note that $P_\sigma(x)$ has value ∞ on its boundary, then $x(\sigma)$ must be an interior point.

Similar to Lemma 10.2.1 and Lemma 10.2.2, we can prove the following results.

Lemma 10.3.1 *Let $\sigma_2 > \sigma_1 > 0$, then*

$$f(x(\sigma_2)) \leq f(x(\sigma_1)), \quad (10.3.9)$$

$$\sum_{i=1}^m h(c_i(x(\sigma_2))) \geq \sum_{i=1}^m h(c_i(x(\sigma_1))). \quad (10.3.10)$$

Lemma 10.3.2 *Set $\delta = \sum_{i=1}^m h(c_i(x(\sigma)))$. Then $x(\sigma)$ is a solution of problem*

$$\min_{x \in R^n} f(x) \quad (10.3.11)$$

$$\text{s.t.} \quad \sum_{i=1}^m h(c_i(x)) \leq \delta. \quad (10.3.12)$$

When δ is sufficiently large, the problem (10.3.11)-(10.3.12) can be regarded as an approximation to

$$\min_{x \in R^n} f(x) \quad (10.3.13)$$

$$\text{s.t.} \quad \sum_{i=1}^m h(c_i(x)) < +\infty. \quad (10.3.14)$$

By the definition of $h(c_i)$, (10.3.14) is equivalent to

$$c_i(x) > 0, \quad i = 1, \dots, m. \quad (10.3.15)$$

The difference between (10.3.15) and (10.3.2) is whether the boundary of the feasible region is feasible points or not. If $\sigma > 0$ is very large and δ in Lemma

10.3.2 is very large, then $x(\sigma)$ is close to the boundary of the feasible region of (10.3.2). Hence, the feasible region of (10.3.12) is also close to that of the original problem. Therefore $x(\sigma)$ is close to the solution of the original problem.

If $\sigma > 0$ is very large but δ bounded, it follows from the definition of (10.3.4) that the interior point penalty function $P_\sigma(x)$ is very close to $f(x)$ near $x(\sigma)$. In this case, $x(\sigma)$ is regarded approximately as a local minimizer of $f(x)$. Based on these analyses, an algorithm can be written as follows. We assume that $h(\cdot)$ satisfies (10.3.7).

Algorithm 10.3.3 (*Algorithm based on interior point penalty function*)

Step 1. Given x_1 satisfying (10.3.15). Let $\sigma_1 > 0$, $\epsilon \geq 0$, $k := 1$.

Step 2. Starting with x_k solve the problem (10.3.8) for $x(\sigma_k)$. Set $x_{k+1} = x(\sigma_k)$.

Step 3. If

$$\frac{1}{\sigma_k} \sum_{i=1}^m h(c_i(x_{k+1})) \leq \epsilon, \quad (10.3.16)$$

stop; otherwise, set $\sigma_{k+1} = 10\sigma_k$, $k := k + 1$; go to Step 2.

□

For Algorithm 10.3.3, we will establish the following convergence theorem.

Theorem 10.3.4 *Let $f(x)$ be bounded below on the feasible region X . Then Algorithm 10.3.3 will terminate finitely at $\epsilon > 0$, and when it does not terminate finitely, we have that*

$$\lim_{k \rightarrow \infty} \frac{1}{\sigma_k} h(c_i(x_{k+1})) = 0 \quad (10.3.17)$$

and

$$\lim_{k \rightarrow \infty} f(x_k) = \inf_{x \in \text{int}(X)} f(x) \quad (10.3.18)$$

hold, where $\text{int}(X)$ is defined by (10.3.3). Furthermore, any accumulation point of $\{x_k\}$ is the solution of problem (10.3.1)-(10.3.2).

Proof. Obviously, we only need to prove that (10.3.17)-(10.3.18) hold when the algorithm does not terminate finitely.

First, for any $\eta > 0$, there exists $x_\eta \in \text{int}(X)$ such that

$$f(x_\eta) < \inf_{x \in \text{int}(X)} f(x) + \frac{\eta}{2}. \quad (10.3.19)$$

Since the algorithm does not terminate finitely, there is $\sigma_k \rightarrow +\infty$. Hence there exists $\bar{k}$ such that

$$\sigma_k > \frac{2}{\eta} \sum_{i=1}^m h(c_i(x_\eta)), \quad \forall k \geq \bar{k}. \quad (10.3.20)$$

Then, by using the definition of x_{k+1} , and (10.3.19)-(10.3.20), we have

$$P_{\sigma_k}(x_{k+1}) = P_{\sigma_k}(x(\sigma_k)) \leq P_{\sigma_k}(x_\eta),$$

that is

$$\begin{aligned} \frac{1}{\sigma} \sum_{i=1}^m h(c_i(x_{k+1})) &\leq f(x_\eta) + \frac{1}{\sigma_k} \sum_{i=1}^m h(c_i(x_\eta)) - f(x_{k+1}) \\ &\leq \inf_{x \in \text{int}(X)} f(x) + \frac{1}{2}\eta + \frac{1}{2}\eta - f(x_{k+1}) \\ &\leq \eta \end{aligned} \quad (10.3.21)$$

holds for all $k \geq \bar{k}$. Since $\eta > 0$ is arbitrary, it follows from (10.3.21) that (10.3.17) is true.

From the first row in (10.3.21), we also get

$$\begin{aligned} f(x_{k+1}) &\leq f(x_\eta) + \frac{1}{\sigma_k} \sum_{i=1}^m h(c_i(x_\eta)) \\ &\leq \inf_{x \in \text{int}(X)} f(x) + \eta \end{aligned} \quad (10.3.22)$$

holds for all $k \geq \bar{k}$. Then (10.3.18) holds. $\square$

Suppose that the sequence $\{x_k\}$ generated by Algorithm 10.3.3 converges to x^* . If x^* is a strict interior point, then it follows from

$$\nabla f(x_{k+1}) + \frac{1}{\sigma_k} \sum_{i=1}^m h'(c_i(x_{k+1})) \nabla c_i(x_{k+1}) = 0 \quad (10.3.23)$$

that

$$\|\nabla f(x_{k+1})\| = O\left(\frac{1}{\sigma_k}\right). \quad (10.3.24)$$

If the second-order sufficient condition (i.e., $\nabla^2 f(x^*)$ positive definite) is satisfied, then (10.3.24) is equivalent to

$$\|x_{k+1} - x^*\| = O\left(\frac{1}{\sigma_k}\right). \quad (10.3.25)$$

The above discussion also tells us that the rate of $\|x_{k+1} - x^*\| \rightarrow 0$ is, in general, no quicker than $1/\sigma_k$.

Now, let us consider $x_k \rightarrow x^*$, where x^* is a boundary point of the feasible region of (10.3.2), i.e., there exists i such that $c_i(x^*) = 0$. Let $\nabla c_i(x^*) (i \in I(x^*))$ be linearly independent. Let λ_i^* denote the Lagrange multiplier at x^* . From (10.3.23), we have

$$\lim_{k \rightarrow \infty} -h'(c_i(x_{k+1}))/\sigma_k = \lambda_i^*. \quad (10.3.26)$$

Write

$$\lambda^{(k+1)} = (-h'(c_1(x_{k+1})), \dots, -h'(c_m(x_{k+1})))^T / \sigma_k.$$

Define A^* as a matrix consisting of $\nabla c_i(x^*) (i \in I(x^*))$, $\hat{\lambda}^*$ as a vector consisting of those components λ_i^* of λ^* ($i \in I(x^*)$), and note that the definitions of $\hat{\lambda}^{(k+1)}$ and $\hat{c}(x_{k+1})$ are similar to $\hat{\lambda}^*$. By (10.3.26), we have

$$|\lambda_i^{(k+1)}| = O\left(\frac{1}{\sigma_k}\right), \quad \forall i \notin I(x^*). \quad (10.3.27)$$

Since the columns of A^* are linearly independent, the above equality gives

$$\|\hat{\lambda}^{(k+1)} - \hat{\lambda}^*\| = O\left(\|x_{k+1} - x^*\| + \frac{1}{\sigma_k}\right). \quad (10.3.28)$$

Note that by (10.3.23), we obtain

$$W^*(x_{k+1} - x^*) - A^*(\hat{\lambda}^{(k+1)} - \hat{\lambda}^*) = o(\|x_{k+1} - x^*\|) + O\left(\frac{1}{\sigma_k}\right). \quad (10.3.29)$$

Also,

$$-(A^*)^T(x_{k+1} - x^*) = -\hat{c}(x_{k+1}) + o(\|x_{k+1} - x^*\|). \quad (10.3.30)$$

Then the above two equalities give

$$\begin{aligned} & \begin{bmatrix} W^* & -A^* \\ -(A^*)^T & 0 \end{bmatrix} \begin{bmatrix} x_{k+1} - x^* \\ \hat{\lambda}^{(k+1)} - \hat{\lambda}^* \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ -\hat{c}(x_{k+1}) \end{bmatrix} + o(\|x_{k+1} - x^*\|) + O\left(\frac{1}{\sigma_k}\right). \end{aligned} \quad (10.3.31)$$

Suppose that $\lambda_i^* > 0$ ($i \in I(x^*)$). Then for an inverse barrier function and a logarithmic barrier function, by using (10.3.26), we obtain respectively

$$c_i(x_{k+1}) = O\left(\frac{1}{\sqrt{\sigma_k}}\right), \quad i \in I(x^*) \quad (10.3.32)$$

and

$$c_i(x_{k+1}) = O\left(\frac{1}{\sigma_k}\right), \quad i \in I(x^*). \quad (10.3.33)$$

Hence, provided that the second-order sufficient condition is satisfied, for the inverse barrier function and logarithmic barrier function, we have

$$\|x_{k+1} - x^*\| = O\left(\frac{1}{\sqrt{\sigma_k}}\right) \quad (10.3.34)$$

and

$$\|x_{k+1} - x^*\| = O\left(\frac{1}{\sigma_k}\right). \quad (10.3.35)$$

From (10.3.34)-(10.3.35), it is easy to see that the logarithmic barrier function converges more quickly than the inverse barrier function does.

Now we consider solving (10.3.8) inexactly by interior point function methods. Suppose that both $f(x)$ and $h(c_i(x))$ are convex functions of x , then $P_\sigma(x)$ is also convex with respect to x . Given starting point x_k , then for problem

$$\min P_{\sigma_k}(x), \quad (10.3.36)$$

the Newton step is

$$d_k = -[\nabla^2 P_{\sigma_k}(x_k)]^{-1} \nabla P_{\sigma_k}(x_k). \quad (10.3.37)$$

To avoid solving the subproblem exactly, the $x_k + d_k$ is regarded as an approximate solution of (10.3.36). For simplicity, we assume

$$h(c_i) = -\log c_i(x). \quad (10.3.38)$$

Set $x_{k+1} = x_k + d_k$. Then we have from (10.3.37) that

$$\nabla^2 P_{\sigma_k}(x_k)(x_{k+1} - x_k) = -\nabla P_{\sigma_k}(x_k). \quad (10.3.39)$$

Note that

$$\begin{aligned} \nabla_x P_{\sigma_k}(x_k) &= \nabla f(x_k) - \sum_{i=1}^m \frac{1}{\sigma_k} \frac{1}{c_i(x_k)} \nabla c_i(x_k), \\ \nabla_x^2 P_{\sigma_k}(x_k) &= \nabla^2 f(x_k) - \sum_{i=1}^m \frac{1}{\sigma_k} \frac{1}{c_i(x_k)} \nabla^2 c_i(x_k) \\ &\quad + \sum_{i=1}^m \frac{1}{\sigma_k} \frac{1}{(c_i(x_k))^2} \nabla c_i(x_k) \nabla c_i(x_k)^T. \end{aligned}$$

Write

$$\lambda_i^{(k)} = \frac{1}{\sigma_k c_i(x_k)}.$$

Then (10.3.39) can be written as

$$\begin{aligned} &\left(\nabla^2 f(x_k) - \sum_{i=1}^m \lambda_i^{(k)} \nabla^2 c_i(x_k) + \sum_{i=1}^m \lambda_i^{(k)} \frac{1}{c_i(x_k)} \nabla c_i(x_k) \nabla c_i(x_k)^T \right) \\ &\cdot (x_{k+1} - x_k) = - \left[\nabla f(x_k) - \sum_{i=1}^m \lambda_i^{(k)} \nabla c_i(x_k) \right]. \end{aligned} \quad (10.3.40)$$

If the x_{k+1} defined above lies in the interior of the feasible region, it is regarded as next iterate. Otherwise, there is $\bar{\alpha} > 0$ such that the point $x_k + \bar{\alpha}_k d_k$ lies on the boundary of the feasible region. In such a case, we set

$$x_{k+1} = x_k + 0.9 \bar{\alpha}_k d_k. \quad (10.3.41)$$

So, the x_{k+1} still is an interior point. Hence, an inexact interior point penalty function algorithm for subproblem (10.3.36) can be written as follows.

Algorithm 10.3.5 (*Inexact Log-Barrier Function Method*)

Step 1. Given x_1 satisfying (10.3.15), $\sigma_1 > 0, \epsilon \geq 0, k := 1$.

Step 2. Compute $\lambda_i^{(k)} = \frac{1}{\sigma_k c_i(x_k)}$, $i = 1, \dots, m$;

$$d_k = -[\nabla^2 f(x_k) - \sum_{i=1}^m \lambda_i^{(k)} \nabla^2 c_i(x_k)]$$

$$\begin{aligned}
& + \sum_{i=1}^m \lambda_i^{(k)} \frac{1}{c_i(x_k)} \nabla c_i(x_k) \nabla c_i(x_k)^T]^{-1} \\
& \cdot (\nabla f(x_k) - \sum_{i=1}^m \lambda_i^{(k)} \nabla c_i(x_k)); \quad (10.3.42)
\end{aligned}$$

If $d_k \neq 0$ go to Step 3;
 If $\|\nabla f(x_k)\| \leq \epsilon$, stop;
 Otherwise, $\sigma_k := 10\sigma_k$; go to step 2.

Step 3. Set $\alpha_k = 1$;
 If $x_k + d_k$ is an interior point, go to Step 4;
 Find $1 \geq \bar{\alpha}_k > 0$ such that $x_k + \bar{\alpha}_k d_k$ is on the boundary of the feasible region;
 Set $\alpha_k := 0.9\bar{\alpha}_k$.

Step 4. Set $x_{k+1} := x_k + \alpha_k d_k$;
 If

$$\frac{1}{\sigma_k} \sum_{i=1}^m \log \left(\frac{1}{c_i(x_k)} \right) \leq \epsilon, \quad (10.3.43)$$

stop; $\sigma_{k+1} := 10\sigma_k$; $k := k + 1$; go to Step 2. $\square$

10.4 Augmented Lagrangian Method

In this section we discuss the augmented Lagrangian method (or the method of multiplier penalty function).

We know from §10.1 that this method is an extension of the quadratic penalty function method. It reduces the possibility of ill-conditioning of the subproblem by introducing Lagrange multiplier estimates. In fact, it is a combination of the Lagrangian function and the quadratic penalty function.

For the case where only equality constraints are presented ($m = m_e$), we rewrite the augmented Lagrangian function as

$$P(x, \lambda, \sigma) = f(x) - \sum_{i=1}^m \lambda_i c_i(x) + \frac{1}{2} \sum_{i=1}^m \sigma_i (c_i(x))^2. \quad (10.4.1)$$

When we differentiate with respect to x , we obtain

$$\nabla_x P(x, \lambda, \sigma) = \nabla f(x) - \sum_{i=1}^m (\lambda_i - \sigma_i c_i(x)) \nabla c_i(x), \quad (10.4.2)$$

which suggests the formula

$$\lambda_i^{(k+1)} = \lambda_i^{(k)} - \sigma_i^{(k)} c_i(x_{k+1}). \quad (10.4.3)$$

Now we consider the general problem (10.1.1)–(10.1.3) by the augmented Lagrangian function. We rewrite the augmented Lagrangian function $P(x, \lambda, \sigma)$ (10.1.21) as follows:

$$\begin{aligned} P(x, \lambda, \sigma) &= f(x) + \sum_{i=1}^{m_e} \left[-\lambda_i c_i(x) + \frac{1}{2} \sigma_i c_i^2(x) \right] \\ &+ \sum_{i=m_e+1}^m \begin{cases} \left[-\lambda_i c_i(x) + \frac{1}{2} \sigma_i c_i^2(x) \right], & \text{if } c_i(x) < \frac{\lambda_i}{\sigma_i} \\ -\frac{1}{2} \lambda_i^2 / \sigma_i, & \text{otherwise} \end{cases} \end{aligned} \quad (10.4.4)$$

where λ_i ($i = 1, \dots, m$) are Lagrange multipliers, σ_i ($i = 1, \dots, m$) are penalty parameters.

Consider the k -th iteration, using $\lambda_i^{(k)}$ and $\sigma_i^{(k)}$ to denote corresponding components of λ and σ respectively at the k -th iteration. Let x_{k+1} be the solution of the subproblem

$$\min_{x \in R^n} P(x, \lambda^{(k)}, \sigma^{(k)}). \quad (10.4.5)$$

Then we have

$$\begin{aligned} \nabla f(x_{k+1}) &= \sum_{i=1}^{m_e} [\lambda_i^{(k)} - \sigma_i^{(k)} c_i(x_{k+1})] \nabla c_i(x_{k+1}) \\ &+ \sum_{i=m_e+1}^m \max\{\lambda_i^{(k)} - \sigma_i^{(k)} c_i(x_{k+1}), 0\} \nabla c_i(x_{k+1}). \end{aligned} \quad (10.4.6)$$

Hence we take

$$\lambda_i^{(k+1)} = \lambda_i^{(k)} - \sigma_i^{(k)} c_i(x_{k+1}), \quad i = 1, \dots, m_e; \quad (10.4.7)$$

$$\lambda_i^{(k+1)} = \max\{\lambda_i^{(k)} - \sigma_i^{(k)} c_i(x_{k+1}), 0\}, \quad i = m_e + 1, \dots, m, \quad (10.4.8)$$

as next Lagrange multipliers. By (10.4.6)–(10.4.8), we have that

$$\nabla f(x_{k+1}) - \sum_{i=1}^m \lambda_i^{(k+1)} \nabla c_i(x_{k+1}) = 0, \quad (10.4.9)$$

which indicates that for any $k \geq 2$, the error of the KKT condition for (x_k, λ_k) is

$$\|\nabla_x L(x_k, \lambda^{(k)})\| + \|c^{(-)}(x_k)\| = \|c^{(-)}(x_k)\|, \quad (10.4.10)$$

where $L(x, \lambda)$ is the Lagrangian function defined in (8.2.18), $c^{(-)}(x)$ is a constraint violation function defined in (10.1.5)–(10.1.6). Therefore, for $k \geq 2$, provided that the inequality

$$|c_i^{(-)}(x_{k+1})| \leq \frac{1}{4}|c_i^{(-)}(x_k)| \quad (10.4.11)$$

is not satisfied, we enlarge the penalty parameters, i.e., set

$$\sigma_i^{(k+1)} = 10\sigma_i^{(k)}. \quad (10.4.12)$$

Below, we give an algorithm based on the augmented Lagrangian function.

Algorithm 10.4.1 (*Augmented Lagrangian Method*)

Step 1. Given starting point $x_1 \in R^n$, $\lambda^{(1)} \in R^m$ with $\lambda_i^{(1)} \geq 0$ ($i \in I$); $\sigma_i^{(1)} > 0$ ($i = 1, \dots, m$); $\epsilon \geq 0$, $k := 1$.

*Step 2. Find approximate minimizer x_{k+1} to (10.4.5).
If $\|c^{(-)}(x_{k+1})\|_\infty \leq \epsilon$, stop.*

Step 3. For $i = 1, \dots, m$, set

$$\sigma_i^{(k+1)} = \begin{cases} \sigma_i^{(k)}, & \text{if (10.4.11) holds;} \\ \max[10\sigma_i^{(k)}, k^2], & \text{otherwise.} \end{cases} \quad (10.4.13)$$

Step 4. Update Lagrange multipliers using (10.4.7)–(10.4.8) to obtain $\lambda^{(k+1)}$, $k := k + 1$, go to Step 2. $\square$

A practical implementation of the above algorithm is given in LANCELOT due to Conn, Gould, and Toint [68].

Now we establish the finite termination of Algorithm 10.4.1.

Theorem 10.4.2 *Let the feasible region X of problem (10.1.1)–(10.1.3) be nonempty. Then for some $\epsilon > 0$, Algorithm 10.4.1 is either finitely terminated, or the sequence $\{x_k\}$ produced by Algorithm 10.4.1 satisfies*

$$\liminf_{k \rightarrow \infty} f(x_k) = -\infty. \quad (10.4.14)$$

Proof. Suppose, on the contrary, that the theorem is not true, that is, for some $\epsilon > 0$, Algorithm 10.4.1 does not terminate finitely, and $\{f(x_k)\}$ is bounded below. Define the set J by

$$J = \left\{ i \mid \lim_{k \rightarrow \infty} |c_i^{(-)}(x_k)| = 0, 1 \leq i \leq m \right\}. \quad (10.4.15)$$

Since the algorithm does not terminate finitely, then the set

$$\hat{J} = \{1, 2, \dots, m\} / J \quad (10.4.16)$$

is not empty. Thus, by the construction of the algorithm, for any $i \in \hat{J}$, we have

$$\lim_{k \rightarrow \infty} \sigma_i^{(k)} = +\infty. \quad (10.4.17)$$

Define

$$\mu_i^{(k)} = \lambda_i^{(k)} / \sqrt{\sigma_i^{(k)}}. \quad (10.4.18)$$

It is not difficult to prove that

$$\begin{aligned} \|\mu^{(k+1)}\|_2^2 &\leq \sum_{i=1}^m [\lambda_i^{(k+1)}]^2 / \sigma_i^{(k)} \\ &\leq \|\mu^{(k)}\|_2^2 + 2[P(x_{k+1}, \lambda^{(k)}, \sigma^{(k)}) - f(x_{k+1})] \\ &\quad - 2[P(\bar{x}, \lambda^{(k)}, \sigma^{(k)}) - f(\bar{x})] \\ &\leq \|\mu^{(k)}\|_2^2 + 2[f(\bar{x}) - f(x_{k+1})], \end{aligned} \quad (10.4.19)$$

where $\bar{x}$ is any feasible point of (10.1.1)–(10.1.3). Since $\{f(x_k)\}$ is bounded below, then (10.4.19) suggests that there exists $\delta > 0$ such that

$$\|\mu^{(k)}\|_2^2 \leq \delta k \quad (10.4.20)$$

holds for all k . Set

$$\tilde{J} = \{i \mid \lim_{k \rightarrow \infty} \sigma_i^{(k)} = +\infty\}. \quad (10.4.21)$$

Equation (10.4.17) indicates that $\hat{J} \subseteq \tilde{J}$. By the definition of x_{k+1} , we have

$$\begin{aligned} &f(\bar{x}) + \sum_{i > m_e} \frac{1}{2} \sigma_i^{(k)} \left[\left(c_i(\bar{x}) - \frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)_-^2 - \left(\frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 \right] \\ &\geq f(x_{k+1}) + \sum_{i \leq m_e} \frac{1}{2} \sigma_i^{(k)} \left[\left(c_i(x_{k+1}) - \frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 - \left(\frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 \right] \\ &\quad + \sum_{i > m_e} \frac{1}{2} \sigma_i^{(k)} \left[\left(c_i(x_{k+1}) - \frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)_-^2 - \left(\frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 \right], \end{aligned} \quad (10.4.22)$$

where $\bar{x}$ is any feasible point of (10.1.1)–(10.1.3), $(\alpha)_-$ denotes $\min\{0, \alpha\}$. By use of (10.4.20)–(10.4.22), we can deduce that

$$\begin{aligned} f(\bar{x}) - f(x_{k+1}) &\geq O(k) \\ &+ \sum_{\substack{i \leq m_e \\ i \in \tilde{J}}} \frac{1}{2} \sigma_i^{(k)} \left[\left(c_i(x_{k+1}) - \frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 - \left(\frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 \right] \\ &+ \sum_{\substack{i > m_e \\ i \in \tilde{J}}} \frac{1}{2} \sigma_i^{(k)} \left[\left(c_i(x_{k+1}) - \frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 - \left(\frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)^2 \right]. \end{aligned} \quad (10.4.23)$$

Since the algorithm does not terminate finitely, then for any k , there exists $\bar{k} > k$ such that for some $i \in \tilde{J}$, we have $\sigma_i^{(\bar{k}+1)} > \sigma_i^{(\bar{k})}$ and

$$|c_i(x_{\bar{k}+1})| > \epsilon \text{ if } i \leq m_e$$

or

$$|(c_i(x_{\bar{k}+1}))_-| > \epsilon \text{ if } i > m_e.$$

Then it follows from (10.4.23) that

$$\begin{aligned} f(\bar{x}) - f(x_{\bar{k}+1}) &\geq O(\bar{k}) + \frac{1}{2} \sigma_i^{(\bar{k})} \epsilon^2 + o(\sigma_i^{(\bar{k})}) \\ &\geq O(\bar{k}) + \frac{1}{4} \bar{k}^2 \epsilon \end{aligned} \quad (10.4.24)$$

$$\geq \frac{1}{8} \bar{k}^2 \epsilon. \quad (10.4.25)$$

This contradicts the fact that the $\{f(x_k)\}$ is bounded below. The contradiction proves the theorem. $\square$

Theorem 10.4.3 *Let the feasible region X of problem (10.1.1)–(10.1.3) be nonempty. Then for $\epsilon = 0$, any accumulation point x^* of the sequence $\{x_k\}$ generated by Algorithm 10.4.1 is feasible. Further, if $\{\lambda^{(k)}\}$ is bounded, then x^* is the solution of the original problem (10.1.1)–(10.1.3).*

Proof. By Algorithm 10.4.1 and Theorem 10.4.2, we have

$$\lim_{k \rightarrow \infty} \|c^-(x_k)\| = 0. \quad (10.4.26)$$

Hence, any accumulation point of $\{x_k\}$ is a feasible point of (10.1.1)–(10.1.3).

Suppose that $\{\lambda^{(k)}\}$ is bounded for all k . Then by (10.4.22) and (10.4.26), we can deduce that

$$\begin{aligned} f(\bar{x}) &\geq f(x_{k+1}) + \sum_{i \leq m_e} \frac{1}{2} \sigma_i^{(k)} c_i^2(x_{k+1}) \\ &\quad + \sum_{i > m_e} \frac{1}{2} \sigma_i^{(k)} \left[\left(c(x_{k+1}) - \frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)_-^2 - \left(-\frac{\lambda_i^{(k)}}{\sigma_i^{(k)}} \right)_-^2 \right] + o(1) \\ &\geq f(x_{k+1}) + o(1). \end{aligned} \quad (10.4.27)$$

Since $\bar{x} \in X$ is arbitrary, we have

$$\lim_{k \rightarrow \infty} f(x) = \inf_{x \in X} f(x). \quad (10.4.28)$$

Thus, any accumulation point x^* of $\{x_k\}$ is the solution of the original problem. $\square$

Finally, we consider the convergence rate of Algorithm 10.4.1. For simplicity, we consider the case with equality constraints only.

Suppose that $x_k \rightarrow x^*$. By Theorem 10.4.3, x^* is the solution of (10.1.1)–(10.1.3). Let $A(x^*)$ have full column rank with $\text{rank}(A(x^*)) = m$. Let $A(x_k)$ have full column rank for all k too. Then the $\lambda^{(k+1)}$ generated by Algorithm 10.4.1 is equivalent to the $\lambda(x_{k+1})$ defined by the following expression

$$A(x_{k+1})\lambda(x_{k+1}) = g(x_{k+1}), \quad (10.4.29)$$

where $g(x) = \nabla f(x)$. Note that

$$\lambda(x) = [A(x)]^+ g(x). \quad (10.4.30)$$

It is not difficult to get

$$\nabla \lambda(x) = [A(x)]^+ W(x), \quad (10.4.31)$$

where

$$W(x) = \nabla^2 f(x) - \sum_{i=1}^m [\lambda(x)]_i \nabla^2 c_i(x). \quad (10.4.32)$$

By (10.4.7), we have

$$\lambda(x_{k+1}) + D_k c(x_{k+1}) = \lambda(x_k), \quad (10.4.33)$$

where

$$D_k = \begin{bmatrix} \sigma_1^{(k)} & & 0 \\ & \ddots & \\ 0 & & \sigma_m^{(k)} \end{bmatrix}. \quad (10.4.34)$$

Then, it suggests by differentiation that

$$[D_k A(x^*)^T + A(x^*)^+ W(x^*)](x_{k+1} - x^*) \approx A(x^*)^+ W(x^*)(x_k - x^*). \quad (10.4.35)$$

The above expression says that, unless $\sigma^{(k)} \rightarrow +\infty$, the sequence $\{x_k\}$ generated by Algorithm 10.4.1 is, in general, convergent linearly.

A shortcoming of the augmented Lagrangian function is that it is only once continuously differentiable. Hence it is possible that there will be some numerical difficulties in solving the subproblem (10.4.5).

10.5 Smooth Exact Penalty Functions

For the equality-constrained problem

$$\min_{x \in R^n} f(x), \quad (10.5.1)$$

$$\text{s.t.} \quad c(x) = 0, \quad (10.5.2)$$

Fletcher [126] first presented a smooth exact penalty function

$$P(x, \sigma) = f(x) - \lambda(x)^T c(x) + \frac{1}{2} c(x)^T D c(x), \quad (10.5.3)$$

where $\lambda(x)$ is given by (10.1.23), $D = \text{diag}(\sigma_1, \dots, \sigma_m)$. From the discussions in §10.1 and §10.4, we know that, if the second-order sufficient condition holds and σ_i are sufficiently large, then the local minimizer of (10.5.1)–(10.5.2) is a strict local minimizer of the penalty function (10.5.3). Conversely, if $\bar{x}$ is a minimizer of (10.5.3) and $c(\bar{x}) = 0$, then $\bar{x}$ is also the minimizer of problem (10.5.1)–(10.5.2).

If we set all σ_i equal in (10.5.3), then a simple form of Fletcher's smooth exact penalty function

$$P(x, \sigma) = f(x) - \lambda(x)^T c(x) + \frac{1}{2} \sigma \|c(x)\|_2^2 \quad (10.5.4)$$

is obtained. For this penalty function, let $x(\sigma)$ be a solution of the subproblem

$$\min_{x \in R^n} P(x, \sigma). \quad (10.5.5)$$

We have, similar to (10.2.5), that

$$\|c(x(\sigma_2))\|_2 \leq \|c(x(\sigma_1))\|_2, \quad \forall \sigma_2 \geq \sigma_1 > 0. \quad (10.5.6)$$

Comparing with the simple penalty function, by use of (10.5.5) and (10.5.4), we need not require $\sigma \rightarrow +\infty$. Thus it is possible to attempt the solution of original problem (10.5.1)–(10.5.2) by solving (10.5.5) without needing $\sigma \rightarrow +\infty$. In addition, the exact penalty function (10.5.3) is smooth, and thus the convergence rate of methods to solve the unconstrained optimization problem (10.5.5) is rapid. A drawback of this approach is, however, that computing $\nabla_x P(x, \sigma)$ needs computation of $\nabla \lambda(x)$, and further $\nabla^2 f(x)$ and $\nabla^2 c_i(x)$, ($i = 1, \dots, m_e$). It is expensive.

If, in (10.5.3), we replace D by

$$2\sigma A^+(A^+)^T, \quad (10.5.7)$$

where $A = \nabla c(x)$, we obtain

$$P(x) = f(x) - \pi(x)^T c(x), \quad (10.5.8)$$

where

$$\pi(x) = A^+(g(x) - \sigma(A^+)^T c(x)). \quad (10.5.9)$$

It is not difficult to find that $\pi(x)$ is the Lagrange multiplier of the problem

$$\min_{d \in R^n} \quad \frac{1}{2} \sigma d^T d + g(x)^T d \quad (10.5.10)$$

$$\text{s.t.} \quad A(x)^T d + c(x) = 0. \quad (10.5.11)$$

For general inequality constrained optimization problem (10.1.1)–(10.1.3), we can define $\pi(x)$ as the Lagrange multiplier of subproblem

$$\min_{d \in R^n} \quad g(x)^T d + \frac{1}{2} \sigma \|d\|_2^2, \quad (10.5.12)$$

$$\text{s.t.} \quad c_i(x) + d^T \nabla c_i(x) = 0, \quad i \in E, \quad (10.5.13)$$

$$c_i(x) + d^T \nabla c_i(x) \geq 0, \quad i \in I, \quad (10.5.14)$$

and then construct the penalty function

$$P(x) = f(x) - \pi(x)^T c(x). \quad (10.5.15)$$

The multiplier $\pi(x)$ can also be obtained by solve the dual problem of (10.5.12)-(10.5.14)

$$\min_{\substack{\pi_i \geq 0 \\ i \in I}} \frac{1}{2} \left\| g(x) - \sum_{i=1}^m \pi_i \nabla c_i(x) \right\|_2^2 + \sigma \pi^T c(x). \quad (10.5.16)$$

As an alternative to the subproblem (10.5.4), we may consider the smooth exact penalty function

$$\begin{aligned} P(x, \lambda) = & f(x) - c(x)^T \lambda + \frac{1}{2} \sigma \|c(x)\|_2^2 \\ & + \frac{1}{2} \rho \|M(x)[g(x) - A(x)\lambda]\|_2^2 \end{aligned} \quad (10.5.17)$$

to deal with an equality constrained problem, where $M(x)$ can be $A(x)^T$, $A(x)^+$, or an identity matrix. Equation (10.5.17) may be extended to handle an inequality-constrained problem. We refer the reader to Di Pillo, Grippo and Lampariell [104] or Fletcher [132].

10.6 Nonsmooth Exact Penalty Functions

Let $h(c)$ be a convex function defined on R^m with $h(0) = 0$. If there exists a positive constant $\delta > 0$ such that

$$h(c) \geq \delta \|c\|_1 \quad (10.6.1)$$

holds for all $c \in R^m$, then $h(c)$ is called a strong distance function.

For any strong distance function $h(c)$, we say that the penalty function

$$P_{\sigma, h}(x) = f(x) + \sigma h(c^{(-)}(x)) \quad (10.6.2)$$

is a nonsmooth exact penalty function, where $\sigma > 0$ is a penalty parameter and $c^{(-)}(x)$ is a constraint violation function defined in (10.1.5)–(10.1.6).

For nonsmooth exact penalty function (10.6.2), we give the following theorem about necessity.

Theorem 10.6.1 *Let x^* be a local minimizer of constrained optimization problem (10.1.1)–(10.1.3) satisfying, together with the corresponding Lagrange multiplier vector λ^* , the second-order sufficient condition*

$$d^T \nabla_{xx}^2 L(x^*, \lambda^*) d > 0, \quad \forall 0 \neq d \in LFD(x^*, X). \quad (10.6.3)$$

Then, if

$$\sigma\delta > \|\lambda^*\|_\infty, \quad (10.6.4)$$

the vector x^* is a strict local minimizer of penalty function $P_{\sigma,h}(x)$ defined in (10.6.2).

Proof. Let (10.6.4) hold. Suppose, to the contrary, that the theorem is not true. Then there exist x_k ($k = 1, 2, \dots$) such that $x_k \neq x^*$, $x_k \rightarrow x^*$ and

$$P_{\sigma,h}(x_k) \leq P(x^*), \quad \forall k. \quad (10.6.5)$$

The above expression gives

$$f(x_k) + \sigma\delta \|c^{(-)}(x_k)\|_1 \leq f(x^*). \quad (10.6.6)$$

Without loss of generality, we assume that

$$(x_k - x^*)/\|x_k - x^*\| \rightarrow d. \quad (10.6.7)$$

By (10.6.6) and the definition of Lagrange multiplier, we obtain

$$\begin{aligned} & (\sigma\delta - \|\lambda^*\|_\infty) \|c^{(-)}(x_k)\|_1 \\ &= (g(x^*) - A(x^*)\lambda^*)^T (x_k - x^*) + (\sigma\delta - \|\lambda^*\|_\infty) \|c^{(-)}(x_k)\|_1 \\ &= f(x_k) - \sum_{i=1}^m \lambda_i^* c_i(x_k) - f(x^*) - \frac{1}{2} (x_k - x^*)^T \nabla_{xx}^2 L(x^*, \lambda^*) (x_k - x^*) \\ & \quad + (\sigma\delta - \|\lambda^*\|_\infty) \|c^{(-)}(x_k)\|_1 + o(\|x_k - x^*\|^2) \\ &= f(x_k) + \sigma\delta \|c^{(-)}(x_k)\|_1 - f(x^*) - \sum_{i=1}^m (\|\lambda^*\|_\infty |c_i^{(-)}(x_k)| + \lambda_i^* c_i(x_k)) \\ & \quad - \frac{1}{2} d^T \nabla_{xx}^2 L(x^*, \lambda^*) d \|x_k - x^*\|_2^2 + o(\|x_k - x^*\|_2^2) \\ &\leq -\frac{1}{2} d^T \nabla_{xx}^2 L(x^*, \lambda^*) d \|x_k - x^*\|_2^2 + o(\|x_k - x^*\|_2^2). \end{aligned} \quad (10.6.8)$$

By using (10.6.8) and (10.6.4), and taking the limit, we yield

$$\lim_{k \rightarrow \infty} \frac{\|c^{(-)}(x_k)\|_1}{\|x_k - x^*\|} = 0, \quad (10.6.9)$$

which indicates

$$d \in \text{LFD}(x^*, X). \quad (10.6.10)$$

From the second-order sufficient condition we have

$$d^T \nabla_{xx}^2 L(x^*, \lambda^*) d > 0 \quad (10.6.11)$$

which shows that the last row in inequality (10.6.8) is negative when k is sufficiently large. Then it produces a contradiction. The contradiction proves the theorem. $\square$

The common nonsmooth exact penalty functions are the L_1 exact penalty function

$$P_1(x) = f(x) + \sigma \|c^{(-)}(x)\|_1 \quad (10.6.12)$$

and the L_∞ exact penalty function

$$P_\infty(x) = f(x) + \sigma \|c^{(-)}(x)\|_\infty. \quad (10.6.13)$$

For nonsmooth exact penalty function (10.6.2), let $x(\sigma)$ be a minimizer of the problem

$$\min_{x \in R^n} P_{\sigma, h}(x). \quad (10.6.14)$$

Completely similar to Lemma 10.2.1 and Lemma 10.2.2, we have the following lemmas. The proofs are omitted.

Lemma 10.6.2 *Let $\sigma_2 > \sigma_1 > 0$. Then we have*

$$f(x(\sigma_2)) \geq f(x(\sigma_1)), \quad (10.6.15)$$

$$h(c^{(-)}(x(\sigma_2))) \leq h(c^{(-)}(x(\sigma_1))). \quad (10.6.16)$$

Lemma 10.6.3 *Let $\eta = h(c^{(-)}(x(\sigma)))$, then $x(\sigma)$ is also the solution of cconstrained problem*

$$\min_{x \in R^n} f(x) \quad (10.6.17)$$

$$\text{s.t.} \quad h(c^{(-)}(x)) \leq \eta. \quad (10.6.18)$$

It is advantageous for an exact penalty function that it is possible to attempt exactly the solution of a constrained optimization problem by solving only a single or finitely many unconstrained problems.

The nonsmooth exact penalty methods can be written in the following form:

Algorithm 10.6.4 (*Nonsmooth Exact Penalty Method*)

Step 1. Given $x_1 \in R^n, \sigma_1 > 0, k := 1$.

Step 2. Solve

$$\min_{x \in R^n} P_{\sigma, h}(x) \quad (10.6.19)$$

at x_k to obtain $x(\sigma)$.

Step 3. If $c^{(-)}(x(\sigma_k)) = 0$, stop;
 $x_{k+1} := x(\sigma_k), \sigma_{k+1} := 10\sigma_k$;
 $k := k + 1$; go to Step 2. $\square$

Note that since $P_{\sigma, h}(x)$ is an exact penalty function, we can obtain the exact solution of the original problem provided that σ is sufficiently large. The following is the convergence result of Algorithm 10.6.4.

Theorem 10.6.5 *Let $f(x), c_i(x)$ ($i = 1, \dots, m$) be twice continuously differentiable. Let the feasible region of constrained optimization problem (10.1.1)–(10.1.3) be nonempty. If second order sufficient condition (10.6.3) holds, then either Algorithm 10.6.4 terminates at a strict local minimizer of problem (10.1.1)–(10.1.3) in finitely many iterations, or the generated sequence satisfies $\|x_k\| \rightarrow \infty$.*

Proof. (1) If the algorithm terminates finitely at $x(\sigma)$, then $x(\sigma_k)$ must be a local minimizer of $P_{\sigma_k, h}(x)$. By Lemma 10.6.6, which will be presented below, $x(\sigma_k)$ is also a local minimizer of the original problem (10.1.1)–(10.1.3). Since the second-order sufficient condition is satisfied at $x(\sigma_k)$, then $x(\sigma_k)$ is a strict local minimizer.

(2) Now we prove the second conclusion by contradiction. Suppose that the theorem is not true. Then, for any k , we have $c^{(-)}(x_k) \neq 0$, $\{\|x_k\|\}$ has a bounded subsequence and $\sigma_k \rightarrow \infty$. Let $\bar{x}$ be any local minimizer of problem (10.1.1)–(10.1.3). By the definition of $x(\sigma_k)$, we have

$$f(x_{k+1}) + \sigma_k h(c^{(-)}(x_{k+1})) \leq f(\bar{x}) + \sigma_k h(c^{(-)}(\bar{x})) = f(\bar{x}), \quad (10.6.20)$$

which means

$$\sigma_k h(c^{(-)}(x_{k+1})) \leq f(\bar{x}) - f(x_{k+1}).$$

Then we have

$$\lim_{k \rightarrow \infty} h(c^{(-)}(x_{k+1})) = 0. \quad (10.6.21)$$

The above expression and (10.6.1) suggest that

$$\lim_{k \rightarrow \infty} \|c^{(-)}(x_k)\| = 0. \quad (10.6.22)$$

Since $\{\|x_k\|\}$ has a bounded subsequence, we may let $\hat{x}$ be an accumulation point of $\{x_k\}$ and therefore using (10.6.22) we have

$$c^{(-)}(\hat{x}) = 0. \quad (10.6.23)$$

Let $x_{k_j} \rightarrow \hat{x}$. If $\hat{x}$ is not a local minimizer of (10.1.1)–(10.1.3), then there exists $\tilde{x}$ sufficiently approaching $\hat{x}$ and we have

$$f(\tilde{x}) < f(\hat{x}), \quad (10.6.24)$$

$$c^{(-)}(\tilde{x}) = 0. \quad (10.6.25)$$

From (10.6.24) and that $x_{k_j} \rightarrow \hat{x}$, we obtain

$$f(x_{k_j}) > f(\tilde{x}) \quad (10.6.26)$$

for j sufficiently large. Hence, we deduce

$$P_{\sigma_{k_j-1},h}(x_{k_j}) > P_{\sigma_{k_j-1},h}(\tilde{x}), \quad (10.6.27)$$

which contradicts the definition of $\{x_{k_j}\}$. Therefore, $\hat{x}$ is a local minimizer of the original problem (10.1.1)–(10.1.3).

Then it follows from Theorem 10.6.1 that there exist $\bar{\delta}$ and $\bar{\sigma}$ such that

$$P_{\bar{\sigma},h}(x) > P_{\bar{\sigma},h}(\hat{x}), \quad \forall \|x - \hat{x}\| \leq \bar{\delta}, \quad x \neq \hat{x}. \quad (10.6.28)$$

The above expression suggests that

$$P_{\sigma,h}(x) > P_{\sigma,h}(\hat{x}), \quad \forall x \neq \hat{x}, \quad \|x - \hat{x}\| \leq \bar{\delta}, \quad \sigma > \bar{\sigma}. \quad (10.6.29)$$

Since $\sigma_{k_j} \rightarrow \infty$, $x_{k_j} \rightarrow \hat{x}$ and $x_{k_j} \neq \hat{x}$, then there exists j such that $\|x_{k_j} - \hat{x}\| < \bar{\delta}$ and $\sigma_{k_j-1} > \bar{\sigma}$. Hence

$$P_{\sigma_{k_j-1},h}(x_{k_j}) > P_{\sigma_{k_j-1},h}(\hat{x}), \quad (10.6.30)$$

which contradicts the definition of x_{k_j} . The contradiction shows the theorem.

□

If Algorithm 10.6.4 terminates finitely, it is sure that it terminates at a local minimizer of the original problem. This is based on the following lemma.

Lemma 10.6.6 *For any $\sigma > 0$ and $\bar{x} \in R^n$, if $h(c^{(-)}(\bar{x})) = 0$ and $\bar{x}$ is a local minimizer of the nonsmooth exact penalty function $P_{\sigma,h}(x)$, then $\bar{x}$ is also a local minimizer of the constrained optimization problem (10.1.1)–(10.1.3).*

Proof. Let $\bar{x}$ satisfy $h(c^{(-)}(\bar{x})) = 0$ and be a local minimizer of $P_{\sigma,h}(x)$. Suppose, to the contrary, that the lemma is not true. Then there exist x_k , ($k = 1, 2, \dots$), such that $x_k \rightarrow \bar{x}$, $x_k \neq \bar{x}$ and

$$f(x_k) < f(\bar{x}), \quad (10.6.31)$$

$$c^{(-)}(x_k) = 0. \quad (10.6.32)$$

Then, we have

$$P_{\sigma,h}(x_k) < P_{\sigma,h}(\bar{x}), \quad (10.6.33)$$

which contradicts the fact that $\bar{x}$ is a local minimizer of $P_{\sigma,h}(x)$. Then we complete the proof. $\square$

We would like to mention that, in a rare case, it is possible that there is $\|x_k\| \rightarrow \infty$ for Algorithm 10.6.4. For example, consider

$$\min_{x \in R^1} \quad 100e^{-x} - \frac{1}{x^2 + 1} \quad (10.6.34)$$

$$\text{s.t.} \quad xe^{-x} = 0. \quad (10.6.35)$$

Taking $h(c) = |c|$ yields that the penalty function is

$$P_{\sigma}(x) = 100e^{-x} - \frac{1}{x^2 + 1} + \sigma |xe^{-x}|. \quad (10.6.36)$$

For a sufficiently large $\sigma > 0$, the minimizer $x(\sigma)$ of $P_{\sigma}(x)$ satisfies the equation

$$-100 + \frac{2xe^x}{(x^2 + 1)^2} = \sigma(x - 1), \quad (10.6.37)$$

and $x(\sigma) > 1$. Then

$$\lim_{\sigma \rightarrow \infty} \frac{2e^{x(\sigma)}}{(x(\sigma)^2 + 1)^2 \sigma} = 1. \quad (10.6.38)$$

Therefore

$$\lim_{\sigma \rightarrow \infty} x(\sigma) = +\infty, \quad (10.6.39)$$

which says that the sequence generated by Algorithm 10.6.4 satisfies $x_k \rightarrow +\infty$.

We note that when the gradients of the constraint function are linearly dependent, it is possible that the minimizer of the original problem (10.1.1)–(10.1.3) is not a stationary point of the exact penalty function (10.6.2). For example,

$$\min_{x \in R^1} x \quad (10.6.40)$$

$$\text{s.t.} \quad c(x) = x^2 = 0. \quad (10.6.41)$$

Taking $h(c) = c$ yields that, for any given $\sigma > 0$, the solution $x^* = 0$ of problem (10.6.40)–(10.6.41) is not the stationary point of the exact penalty function

$$P_{\sigma,h}(x) = x + \sigma x^2. \quad (10.6.42)$$

However, if the gradients of the constraint function are linearly independent, the minimizer of original problem (10.1.1)–(10.1.3) is also the minimizer of the exact penalty function.

Theorem 10.6.7 *Let x^* be a local minimizer of constrained optimization problem (10.1.1)–(10.1.3) and λ^* be a corresponding Lagrange multiplier. If*

$$\nabla c_i(x^*), i \in E \cup I(x^*) \quad (10.6.43)$$

are linearly independent, then when (10.6.4) holds, the x^ is also a local minimizer of the penalty function (10.6.2).*

Proof. If the theorem is not true, then there exist x_k ($k = 1, 2, \dots$) such that $x_k \neq x^*$, $x_k \rightarrow x^*$ and

$$P_{\sigma,h}(x_k) < P(x^*), \forall k. \quad (10.6.44)$$

Then by (10.6.1) we have

$$f(x_k) + \sigma \delta \|c^{(-)}(x_k)\|_1 < f(x^*). \quad (10.6.45)$$

Similar to the proof of Theorem 10.6.1, we may assume that (10.6.7) holds and use (10.6.8) to get

$$d \in \text{LFD}(x^*, X). \quad (10.6.46)$$

The second-order necessary condition gives

$$d^T \nabla_{xx}^2 L(x^*, \lambda^*) d \geq 0, \quad (10.6.47)$$

from which together with (10.6.8), we can deduce that

$$d^T \nabla_{xx}^2 L(x^*, \lambda^*) d = 0. \quad (10.6.48)$$

Then

$$\|c^{(-)}(x_k)\|_1 = o(\|x_k - x^*\|^2). \quad (10.6.49)$$

Since x^* is a local minimizer of the original problem (10.1.1)–(10.1.3), it follows from (10.6.45) that

$$\|c^{(-)}(x_k)\|_1 > 0 \quad (10.6.50)$$

for all sufficiently large k . Since the gradients of all active constraints are linearly independent, then there is some y_k such that

$$c^{(-)}(y_k) = 0 \quad (10.6.51)$$

and

$$\|y_k - x_k\| = O(\|c^{(-)}(x_k)\|). \quad (10.6.52)$$

By use of the optimality of x^* and (10.6.51), we have

$$f(y_k) \geq f(x^*). \quad (10.6.53)$$

On the other hand, by the KKT condition, (10.6.4) and (10.6.45), we can obtain that

$$\begin{aligned} f(y_k) &= f(x_k) + \nabla f(x^*)^T (y_k - x_k) + o(\|y_k - x_k\|) \\ &= f(x_k) + \sum_{i=1}^m \lambda_i^* (y_k - x_k)^T \nabla c_i(x^*) + o(\|y_k - x_k\|) \\ &\leq f(x_k) + \lambda^* \|\infty\| \|c^{(-)}(x_k)\|_1 + o(\|c^{(-)}(x_k)\|_1) \\ &< f(x_k) + \sigma \delta \|c^{(-)}(x_k)\|_1 \\ &< f(x^*), \end{aligned} \quad (10.6.54)$$

which contradicts (10.6.53). The contradiction proves the theorem. $\square$

It is not difficult to see that the equivalence between the nonsmooth exact penalty function (10.6.2) and the constrained optimization problem is based on (10.6.4). In fact, if the inequality (10.6.4) is not satisfied, then the local minimizer of (10.1.1)–(10.1.3) is not necessarily a stationary point of the penalty function (10.6.2).

Theorem 10.6.8 *Let x^* be a local minimizer of constrained optimization (10.1.1)–(10.1.3) and $\nabla f(x^*) \neq 0$. Write*

$$T = \max_{v \in \partial h(0)} \|v\|. \quad (10.6.55)$$

Then, when

$$\sigma \|\nabla c^{(-)}(x^*)\| < \|\nabla f(x^*)\|/T, \quad (10.6.56)$$

x^ is not the stationary point of the penalty function (10.6.2).*

Proof. Since the subgradient of the penalty function (10.6.2) at x^* is

$$\partial P_{\sigma,h}(x^*) = \nabla f(x^*) + \sigma \nabla c^{(-)}(x^*)^T \partial h(0), \quad (10.6.57)$$

then, by (10.6.56), we have

$$0 \notin \partial P_{\sigma,h}(x^*). \quad (10.6.58)$$

Therefore, x^* is not a stationary point of $P_{\sigma,h}(x)$. $\square$

Exercises

1. Use the Courant penalty function method to solve the problem

$$\begin{aligned} \min \quad & -2x_1 + x_2 \\ \text{s.t.} \quad & x_2 - x_1^2 = 0. \end{aligned}$$

2. Apply the inverse penalty function method to solve the problem

$$\begin{aligned} \min \quad & -x_1^2 - x_2^2 \\ \text{s.t.} \quad & x_1 \leq 8, \\ & x_2 \leq 8, \\ & x_1 + x_2 \geq 1 \end{aligned}$$

with the initial point $(2 \ 2)^T$.

3. Apply the logarithmic barrier function method to solve the problem

$$\begin{aligned} \min \quad & x_1 - x_2 + x_2^2 \\ \text{s.t.} \quad & x_1 \geq 0, \\ & x_2 \geq 0 \end{aligned}$$

with the initial point $(1, 1)^T$.

4. Apply the Augmented Lagrangian function method to solve the problem in the previous exercise, using initial multipliers $\lambda_1 = 1$ and $\lambda_2 = 1$. Compare the performances of the two methods (Logarithmic barrier function method and the Augmented Lagrangian function method).

5. Let $x(\sigma)$ be the solution of

$$\min P(x, \sigma) = f(x) + \frac{1}{\sigma} \sum_{i=1}^m \frac{1}{c_i(x)} \quad (10.6.59)$$

in the interior region $\{x | c_i(x) > 0, i = 1, \dots, m\}$, where $\sigma > 0$ is a parameter. Prove that, as σ increases,

- (1) $P(x(\sigma), \sigma)$ is non-increasing;
- (2) $\sum_{i=1}^m \frac{1}{c_i(x(\sigma))}$ is non-decreasing;
- (3) $f(x(\sigma))$ is non-increasing.

6. Discuss the penalty function (10.1.10) when $h(c) = e^c$.

7. Using the approximation

$$\max\{c_1, \dots, c_m\} \approx \log \left(\sum_{i=1}^m e^{c_i} \right),$$

we can replace the L_∞ penalty function by

$$P_e(x) = f(x) + \sigma \log \left(\sum_{i=1}^m e^{|c_i^-(x)|} \right).$$

Study the properties of the above penalty function $P_e(x)$.

8. Introducing the slack variables for the inequality constraints, we can reformulate (8.1.1)–(8.1.3) as

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & c_i(x) = 0, \quad i = 1, \dots, m_e, \\ & c_i(x) - y_i = 0, \quad i = m_e + 1, \dots, m, \\ & y_i \geq 0, \quad i = m_e + 1, \dots, m. \end{aligned}$$

Compare the augmented Lagrange function for the reformulated problem and (10.4.4).

9. Prove (10.5.6).
10. Prove Lemma 10.6.2.